25MML0008_Edge_Yolov8Video.ipynb

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output_tracking.mp4
yolov8n.pt

[1]

```
pip install ultralytics opencv-python
```

```
Collecting ultralytics
  Downloading ultralytics-8.4.27-py3-none-any.whl.metadata (39 kB)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.10.0+cpu)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.25.0+cpu)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
  Downloading ultralytics_thop-2.0.18-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (26.
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics)
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->ultralytic
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralyti
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.11)
```

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[3]

```
import cv2
from ultralytics import YOLO
from google.colab.patches import cv2_imshow
cv2_imshow(annotated_frame)

# ------------------------------
# Load YOLOv8 Model
# ------------------------------
model = YOLO("yolov8n.pt")   # you can use yolov8s.pt / yolov8m.pt also

# ------------------------------
# Input Video
# ------------------------------
video_path = "/content/5097127-uhd_3840_2160_24fps.mp4"
cap = cv2.VideoCapture(video_path)

# ------------------------------
# Output Video Writer
# ------------------------------
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = cap.get(cv2.CAP_PROP_FPS)

out = cv2.VideoWriter(
    "output_tracking.mp4",
    cv2.VideoWriter_fourcc(*"mp4v"),
```

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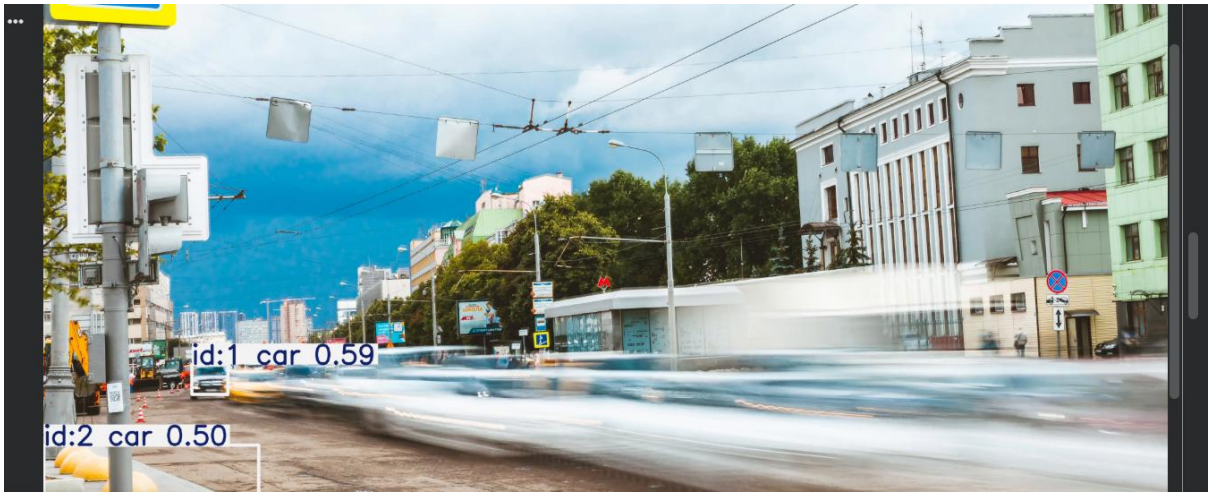
[3]

```
# ------------------------------
# Tracking Loop
# ------------------------------
from google.colab.patches import cv2_imshow

while cap.isOpened():

    ret, frame = cap.read()
    if not ret:
        break

    results = model.track(
        frame,
        persist=True,
        tracker="bytetrack.yaml"
    )

    annotated_frame = results[0].plot()

    # SHOW FRAME (Colab way)
    cv2_imshow(annotated_frame)

    # SAVE
    out.write(annotated_frame)
cap.release()
out.release()
cv2.destroyAllWindows()
```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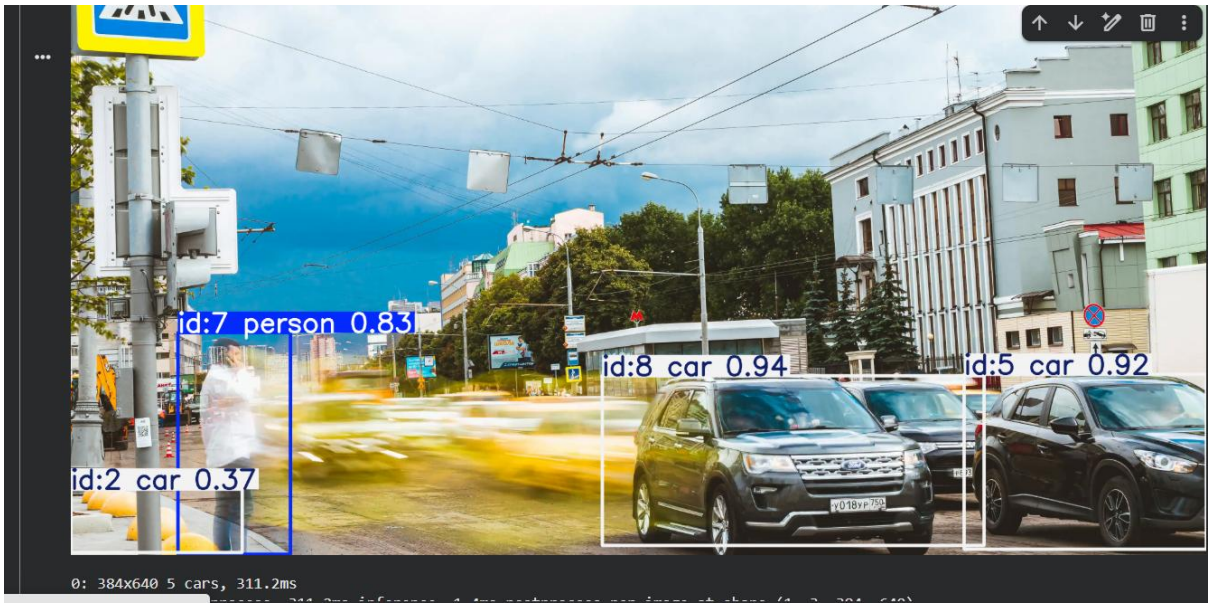
id:2 car 0.50

id:2 car 0.40


id:7 person 0.83
id:8 car 0.94
id:5 car 0.92
id:2 car 0.37
0: 384x640 5 cars, 311.2ms